Euler's Totient Function

$\varphi(n)$

Definition: Euler's Totient Function, $\varphi : \mathbb{N} \to \mathbb{N}$

with

$$\varphi(n) = \mathrm{card}\left(\{ m \in \mathbb{N} \mid 1 \le m < n \mid \gcd(m,n) = 1 \}\right)$$

examples: $\varphi(5) = |\{1,2,3,4\}| = 4$

$\varphi(8) = 4$, $\varphi(15) = 8$

proposition: For a prime p we have $\varphi(p) = p-1$.

proof: Notice the only divisors of p are 1 and p.
Let $1 \le m < p$ then $p \nmid m$ and $2 \le m < p$
doesn't divide p, only $m = 1$. $\Rightarrow \gcd(m,p) = 1$

$\square$

proposition: If $\gcd(m,n) = 1$ then $\varphi(mn) = \varphi(m) \cdot \varphi(n)$.

example: $15 = 3 \cdot 5$ $\quad \varphi(15) = 8$, $\varphi(3) = 2$, $\varphi(5) = 4$

1	2	3
4	5	6
7	8	9
10	11	12
13	14	15

obs: There are $\overset{=2}{\varphi(3)}$ columns with relatively primes to 3. In each of these columns there are $\overset{=4}{\varphi(5)}$ rel. prime to 5.

$$\varphi(15) = \varphi(3) \cdot \varphi(5) = 8$$

proof: Consider

$$\begin{array}{c|c|c|c|c} 1 & 2 & \dots & m-1 & m \\ m+1 & m+2 & \dots & 2m-1 & 2m \\ \vdots & & & & \\ (n-1)m+1 & (n-1)m+2 & \dots & nm-1 & nm \\ \parallel & \parallel & & & \\ nm-m+1 & nm-m+2 & & & \end{array}$$

observe: there are $\varphi(m)$ columns made completely of numbers relatively prime to m.

Say we are in column k: $k, m+k, \dots, nm-m+k$

claim: each of these columns contains exactly $\varphi(n)$ numbers relatively prime to n.

Take 2 numbers from column k, say $lm+k$ and $l'm+k$

note: $lm+k \equiv (l'm+k) \mod n$, because

$\Leftrightarrow (l-l')m \equiv 0 \mod n$

$\Rightarrow n \mid l-l' \Rightarrow l = l'$, cause $l-l' < n$

$\Rightarrow$ each column contains $\varphi(n)$ numbers relatively prime to n.

$\rightarrow \varphi(nm) = \varphi(n)\varphi(m)$ $\square$

(Short) Version for Primes

proposition: Let p, q primes $\Rightarrow \varphi(p \cdot q) = \varphi(p) \cdot \varphi(q)$

note:

$$\begin{array}{cc} 1 \cdot p & 1 \cdot q \\ 2 \cdot p & 2 \cdot q \\ \vdots & \vdots \\ (q-1)p & (p-1) \cdot q \\ q \cdot p & p \cdot q \end{array}$$

$\rightarrow$ cause p, q primes there are $q-1$ numbers divisible by p and $p-1$ different numbers divisible by q.

Further $p \cdot q$ is divisible by p and q. So there are $p \cdot q$ numbers of which $pq - (q-1+p-1+1)$ are relatively prime to p or q:

$$pq-(q+p-1) = pq-q-p+1 = (p-1)(q-1)$$

$$= \varphi(p) \cdot \varphi(q)$$ $\square$

proposition: For a prime p, we have $\varphi(p^r) = p^r - p^{r-1}$

examples: $2^3 \to \varphi(8) = 4$

$$\begin{array}{cc} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{array}$$

relativly prime to 2

multiples of 2

$3^2 \to \varphi(9) = 6$

$$3\left\{\begin{array}{ccc} \overbrace{1 \quad 2}^{2} & & 3 \\ 4 & 5 & 6 \\ \underbrace{7 \quad 8}_{} & & 9 \end{array}\right.$$

rel. prime to 3

multiples of 3

proof: Arrange $1, 2, 3, \dots, p^r - 1, p^r$ in an array:

$$p^{r-1}\left\{\begin{array}{cccc|c} 1 & 2 & 3 & \dots & p \\ p+1 & p+2 & \dots & & 2p \\ 2p+1 & 2p+2 & \dots & & 3p \\ \vdots & \vdots & & & \vdots \\ p^{r-1}+1 & p^{r-1}+2 & \dots & & p^r \end{array}\right.$$

relativly prime to p

not relativly prime to p

$$\underbrace{\qquad\qquad\qquad}_{p-1}$$

$$\Rightarrow \varphi(p^r) = p^{r-1} \cdot (p-1) = p^r - p^{r-1} \qquad \square$$

Theorem: If $n = p_1^{r_1} \cdots p_k^{r_k}$ then

$$\varphi(n) = n\left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right)$$

proof:

$$\begin{aligned} \varphi(n) &= \varphi(p_1^{r_1} \cdots p_k^{r_k}) = \varphi(p_1^{r_1}) \cdot \ldots \cdot \varphi(p_k^{r_k}) \\ &= (p_1^{r_1} - p_1^{r_1 - 1}) \cdot \ldots \cdot (p_k^{r_k} - p_k^{r_k - 1}) \\ &= p_1^{r_1}\left(1 - \frac{1}{p_1}\right) p_2^{r_2}\left(1 - \frac{1}{p_2}\right) \cdots p_k^{r_k}\left(1 - \frac{1}{p_k}\right) \\ &= p_1^{r_1} \cdots p_k^{r_k}\left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right) \\ &= n \cdot \left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right) \qquad \square \end{aligned}$$

Bem: Man muss nur die Primfaktoren von n kennen, aber nicht die Anzahl mit der sie vorkommen.